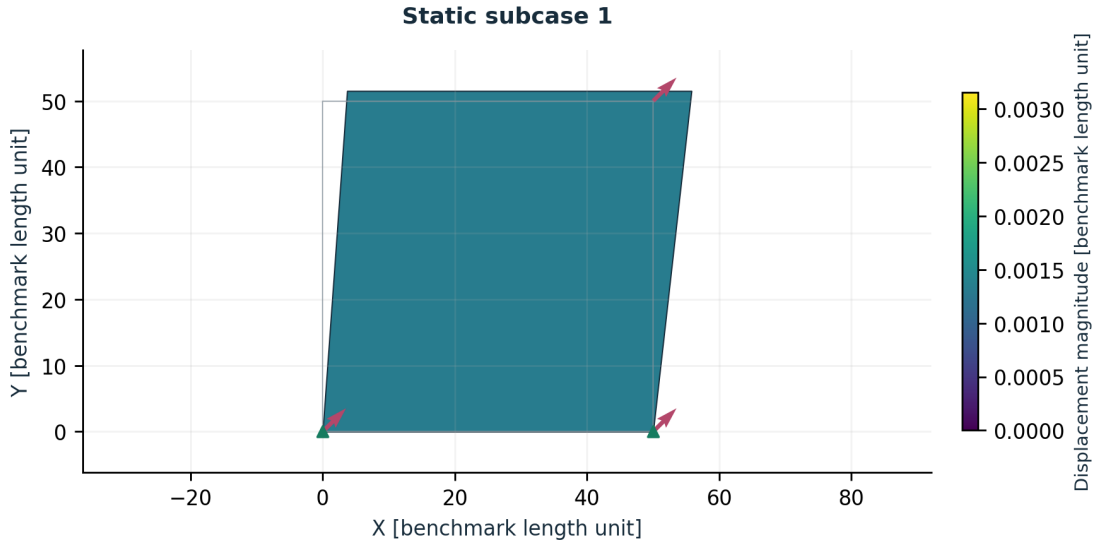
Open-source quadrilateral load patch

MYSTRAN_sol101_simple | SOL 101 | Reproducibility / historical regression

Single 50x50 quadrilateral shell; t=0.1, E=1e7, nu=0.25.

Geometry / mesh	4 nodes; 1 elements. Single 50x50 quadrilateral shell; t=0.1, E=1e7, nu=0.25.
Material / section	E=1e+07, nu=0.25; density=0.1; shell thickness=0.1. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Edge 1011-1012 clamped; three subcases: nodal forces, uniform PLOAD2, uniform PLOAD4.
Applied load	Three distinct subcases: nodal forces, PLOAD2 uniform pressure and PLOAD4 uniform pressure; see explicit deck values.



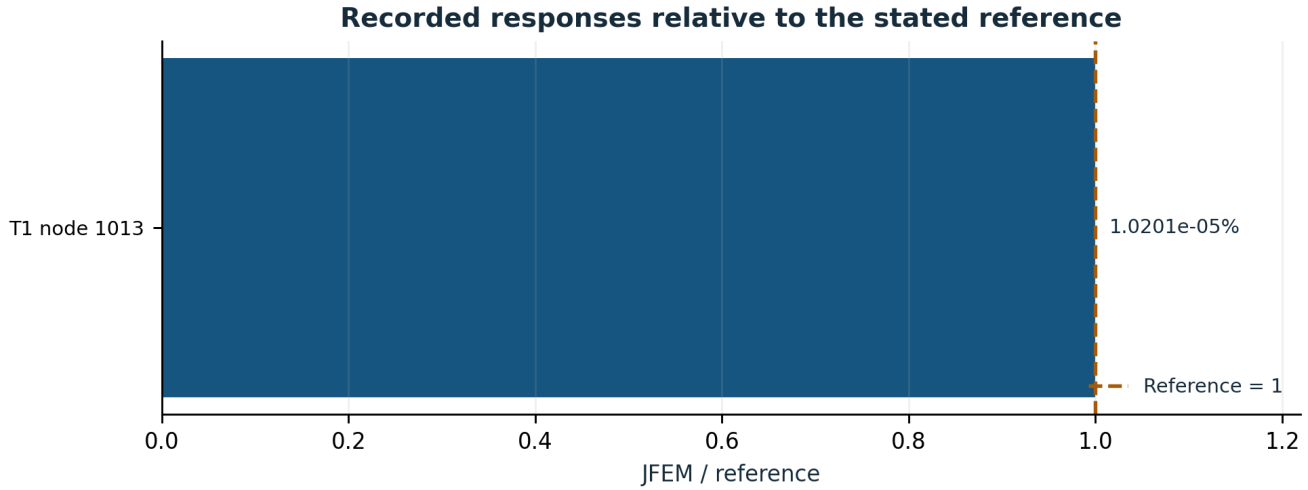
Shape amplified x1.9e+03; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
T1 node 1013 [benchmark length unit]	0.0030533	0.0030533	1.0201e-05%	REGRESSION

Evidence, convergence and interpretation



Stored historical scalar has no independently reproducible public source. This comparison is a regression check and is not independent scientific validation.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MYSTRAN open-source test input provenance (MIT license); stored historical output values are regression targets only. https://github.com/MYSTRANsolver/MYSTRAN

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv